

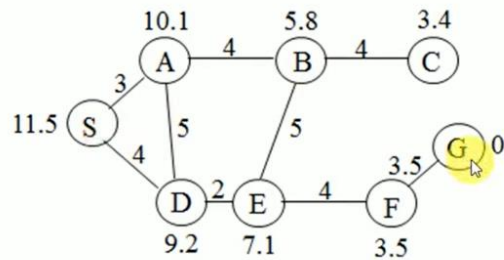
A* (Star) Search in Artificial Intelligence

- Algorithm A* (Hart et al., 1968):

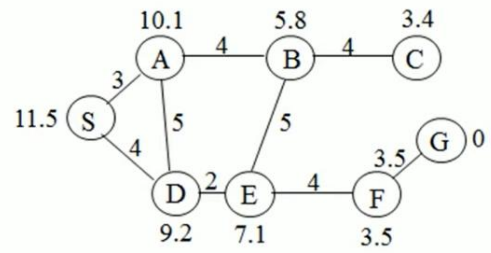
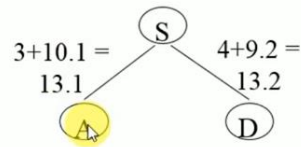
$$f(n) = g(n) + h(n)$$

- $h(n)$ = cost of the cheapest path from node n to a goal state.
- $g(n)$ = cost of the cheapest path from the initial state to node n .

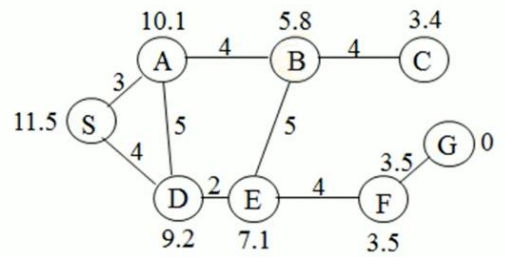
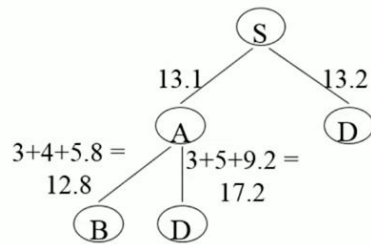
A* (Star) Search in Artificial Intelligence



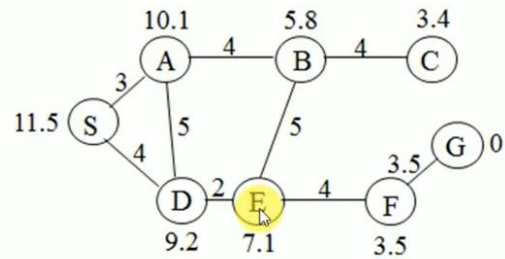
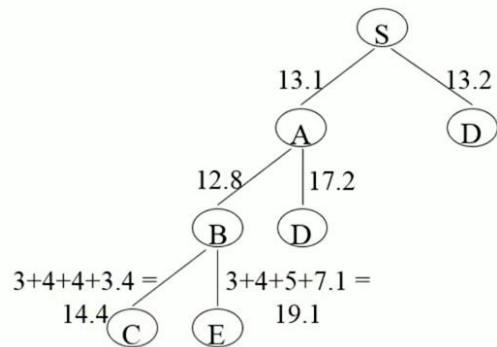
A* (Star) Search in Artificial Intelligence



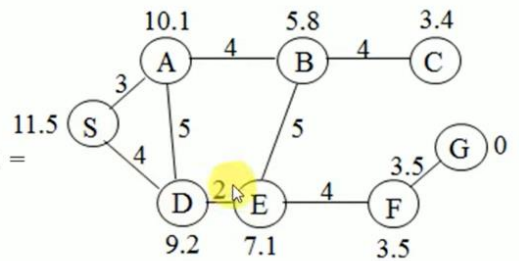
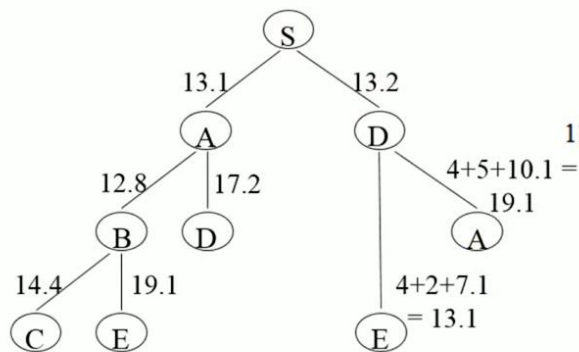
A* (Star) Search in Artificial Intelligence



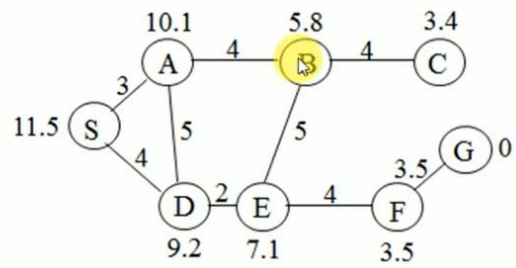
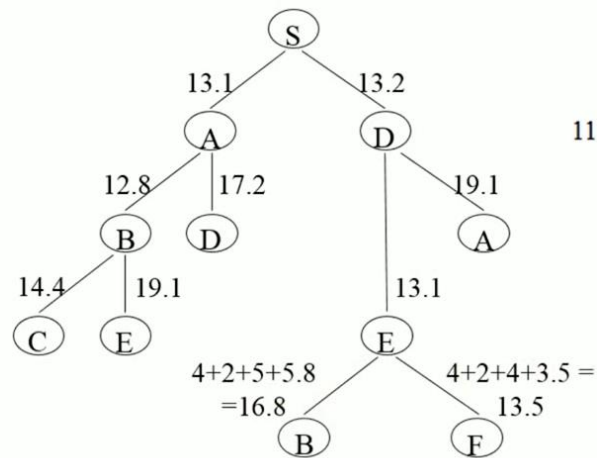
A* (Star) Search in Artificial Intelligence



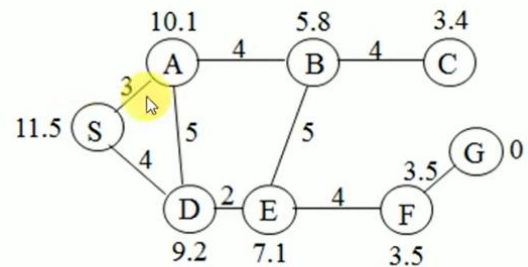
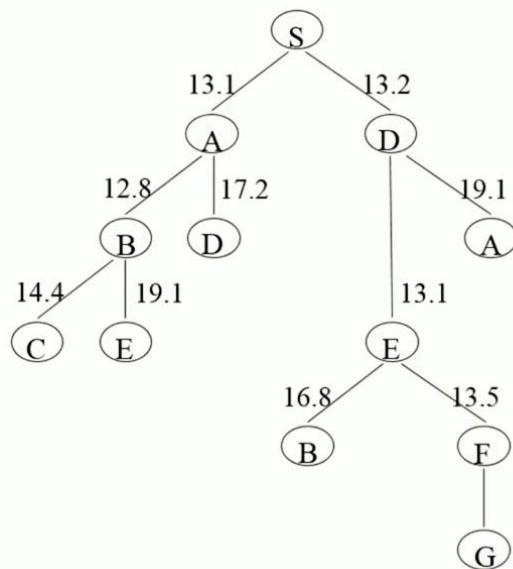
A* (Star) Search in Artificial Intelligence



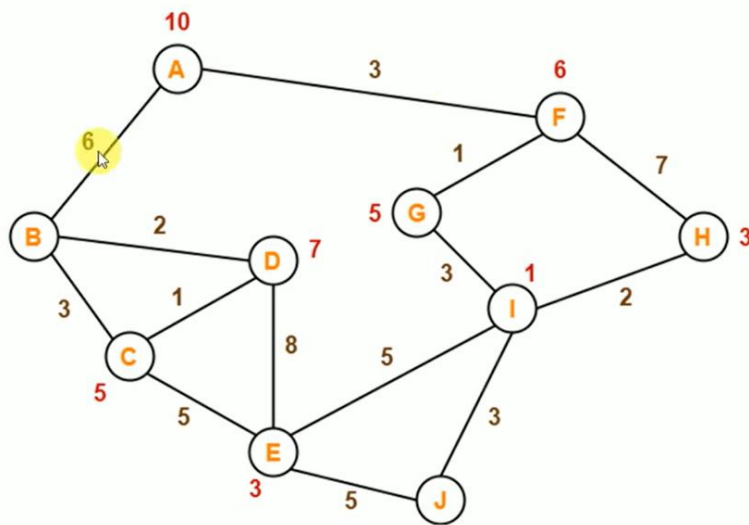
A* (Star) Search in Artificial Intelligence



A* (Star) Search in Artificial Intelligence

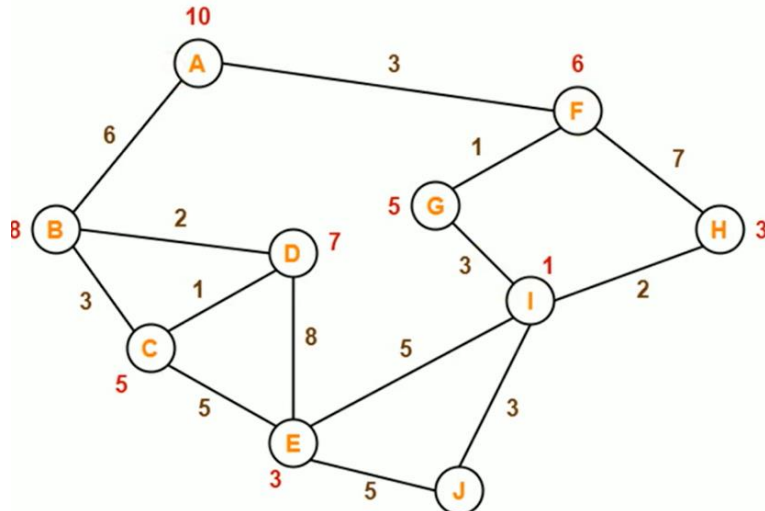


A* Search Algorithm – Numerical Example – 2



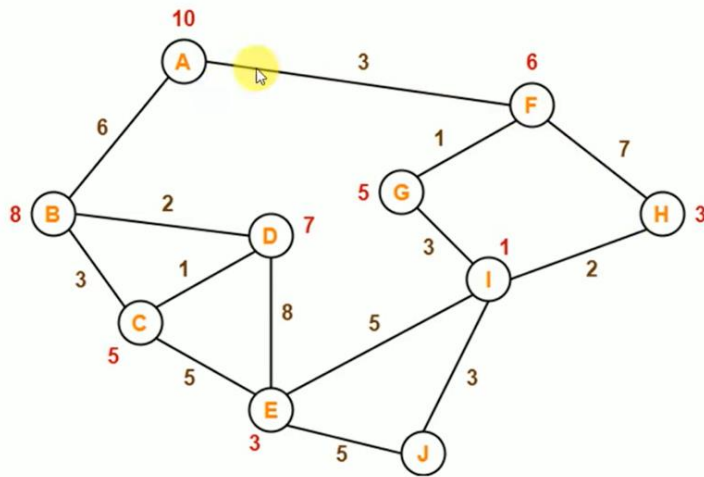
- The numbers written on edges represent the distance between the nodes.

A* Search Algorithm – Numerical Example – 2



- The numbers written on edges represent the distance between the nodes.
- The numbers written on nodes represent the heuristic value.
- Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.

A* Search Algorithm – Numerical Example – 2

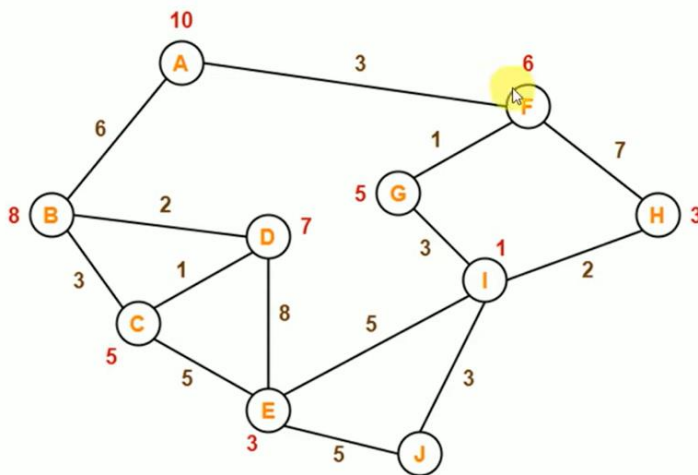


Step-01:

- We start with node A.
- Node B and Node F can be reached from node A.
- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$
 - $f(B) = g(B) + h(B) = 6 + 8 = 14$
 - $f(F) = g(F) + h(F) = 3 + 6 = 9$

A* Search Algorithm – Numerical Example – 2

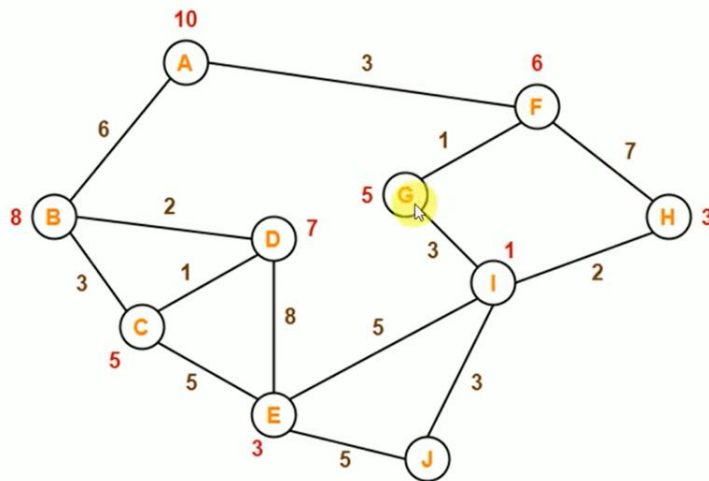


Step-01:

- We start with node A.
- Node B and Node F can be reached from node A.
- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$
 - $f(B) = g(B) + h(B) = 6 + 8 = 14$
 - $f(F) = g(F) + h(F) = 3 + 6 = 9$
 - Since $f(F) < f(B)$, so it decides to go to node F.
- Path- A → F

A* Search Algorithm – Numerical Example – 2

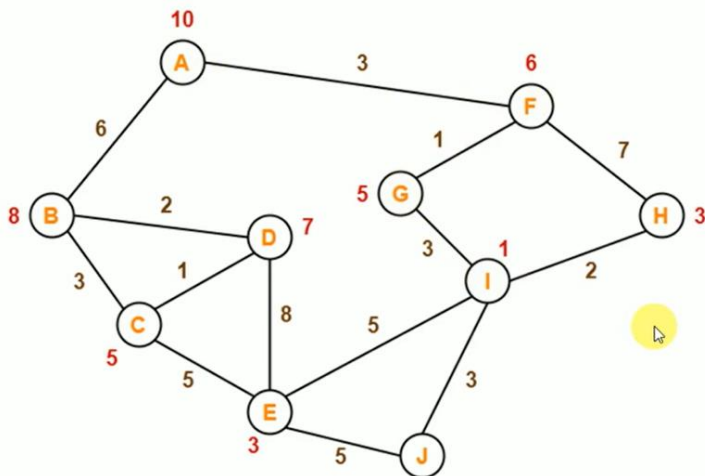


Step-02:

- Node G and Node H can be reached from node F.
- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$
 - $f(G) = g(G) + h(G) = (3+1) + 5 = 9$
 - $f(H) = g(H) + h(H) = (3+7) + 3 = 13$
- Since $f(G) < f(H)$, so it decides to go to node G.
- Path - A → F → G

A* Search Algorithm – Numerical Example – 2



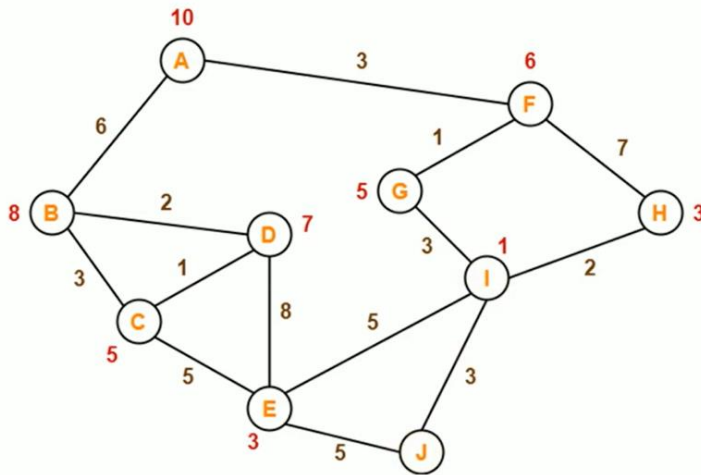
Step-03:

- Node I can be reached from node G.
- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$
 - $f(I) = g(I) + h(I)$

$$= (3+1+3) + 1 = 8$$
- So it decides to go to node I.
- Path - A → F → G → I

A* Search Algorithm – Numerical Example – 2



Step-04:

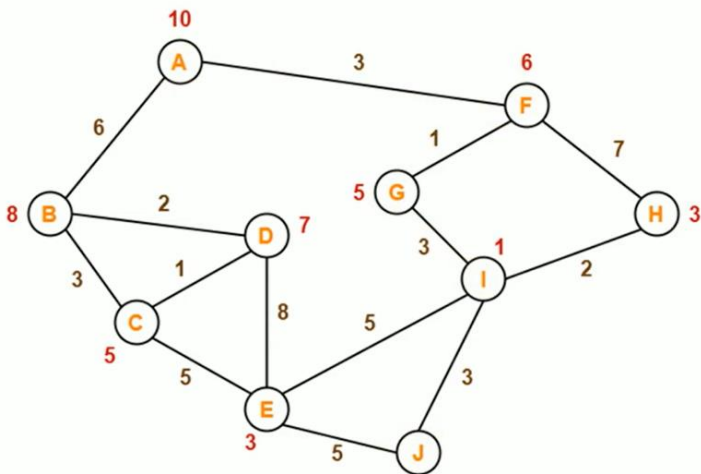
- Node E, Node H and Node J can be reached from node I.

- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$

- $f(E) = (3+1+3+5) + 3 = 15$
- $f(H) = (3+1+3+2) + 3 = 12$
- $f(J) = (3+1+3+3) + 0 = 10$

A* Search Algorithm – Numerical Example – 2



Step-04:

- Node E, Node H and Node J can be reached from node I.

- A* Algorithm calculates

$$f(n) = g(n) + h(n)$$

- $f(E) = (3+1+3+5) + 3 = 15$
- $f(H) = (3+1+3+2) + 3 = 12$
- $f(J) = (3+1+3+3) + 0 = 10$
- Since $f(J)$ is least, so it decides to go to node J.
- Path - $A \rightarrow F \rightarrow G \rightarrow I \rightarrow J$